

Holonomy and prunability as the two edges of one spectral sequence

the Zappa–Szép composition holonomy is the transgression, on Ferri’s $K_{2,3}$

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Draft — work in progress; Kodamai grant deliverable (Theory strand). For Rick (skew-brace / H^2 referee), then Neil. The structural results (§3–§9) are proved; the $K_{2,3}$ material of §7 is computed and machine-checked on the fibre edge; §8 sharpens (and in one respect corrects) the fibre-edge reading in the untwisted core — Ferri prunability is a *degree-one* descent datum, not the degree-two fibre class — with the $|L_\lambda| \geq 3$ linear reduction left **open** (§11).

Abstract

A connected groupoid arising from a Zappa–Szép (matched-pair) factorisation carries two proved degree-two obstruction theories: the multiplicative *composition holonomy* $[\omega]$, whose vanishing is exactly the existence of the global factorisation, and Ferri’s *prunability* obstruction, whose vanishing lets the groupoid be pruned to its loop bundle. Ferri asked (Remark 3.20 of his *dynamical skew braces* paper) for a group-theoretic or cohomological recast of prunability. We give one on his own example, and locate both obstructions inside a single object. When the factorisation is normal, the categorical cohomology of the groupoid is the abutment of the Lyndon–Hochschild–Serre spectral sequence of the extension $1 \rightarrow \mathcal{D} \rightarrow \pi \rightarrow \Gamma \rightarrow 1$; its two degree-two edges are exactly the two obstruction theories — the base edge $H^2(\Gamma; M^{\mathcal{D}})$ carries $[\omega]$, the fibre edge $H^2(\mathcal{D}; M)^{\Gamma}$ carries the additive obstruction — and the transgression $d_2(\varphi) = \varphi_*[\omega]$ is the holonomy. Consequently the two faces are never independent summands: a nonzero base class is a differential, not a summand, so the naive direct sum fails precisely when it would be interesting. On the fibre side we compute the fibre-edge class on Ferri’s own $K_{2,3}$: it is connected (so $[\omega] = 0$) yet non-prunable, and the fibre edge carries a nonzero class $[\pi]$, while a prunable Klein control kills both edges. All finite claims are machine-checked. A companion analysis of the untwisted (abelian-star, subgroup-loop) core sharpens the fibre side: Ferri prunability is there a *degree-one* descent (flat-section) datum — the loop bundle being a kernel — logically independent of the *degree-two* star-reassembly (Schreier) class the fibre edge carries, so the $K_{2,3}$ class detects star-nonsplitness rather than prunability directly. An ideal is a kernel; an extension is a reassembly. The general linear reduction of the descent datum for larger loop groups remains open.

1 Introduction

The problem. A directed container, a matched pair of groups, and a Zappa–Szép product are three names for the same move: build a large composite from two interacting factors that act on one another. Whether the composite exists globally, and whether it can be simplified, are the two questions a compositional-correctness theory must answer. For the groupoid G presented by a

(quiver) skew brace factored over a vertex bundle, both questions have been given cohomological answers, but by *different* theories:

- (H) the *composition holonomy* $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ over the orbit base $\mathrm{Sk}_{\mathcal{C}}$, whose vanishing is equivalent to the existence of the global Zappa–Szép factorisation $K = \mathcal{C} \bowtie \mathcal{D}$ [1, 2]; and
- (P) Ferri’s *prunability* obstruction [7], an additive condition on the loop bundle (the loops over each object must form an additive subgroup), whose vanishing lets G be pruned to its loop subgroupoid, and which lives in the direction-functor cohomology $H_{\mathrm{Cat}_B}^2(G) \cong H^2(\pi)$ of the vertex group [9, 4].

These were proved *separate*: $[\omega]$ vanishes identically for a full loop bundle, so it cannot be prunability [3]. Ferri, working from the skew-brace side, asked for the missing structural picture explicitly.

Ferri, Remark 3.20 (paraphrased) [7]. *Is there a group-theoretic or cohomological description of the prunability obstruction, of the kind available for the associated extension classes?*

The gap. “Separate” is not a structure. Knowing that $[\omega]$ and prunability are two different classes leaves open the natural synthesis: are they two independent summands of one categorical cohomology,

$$H_{\mathrm{cat}}^2(G; M) \stackrel{?}{\cong} \underbrace{H^2(\pi; M)}_{\text{prunability}} \oplus \underbrace{H_{\mathrm{base}}^2(\mathrm{Sk}_{\mathcal{C}}; M)}_{[\omega]}, \quad \text{both summands possibly nonzero?} \quad (1)$$

This “product decomposition” is what one would write down first. It is false, and its failure is the point.

The results. We settle (1) in general and answer Ferri’s question on his own example, by putting both obstructions on one page. The structural statement below is proved; the answer to Remark 3.20 is computed on $K_{2,3}$, and §8 sharpens its reading (prunability is a *degree-one* descent datum, not the degree-two fibre class), leaving the $|L_\lambda| \geq 3$ linear reduction open (§11).

Theorem 1 (Structural, §3–§9). *Let G be a connected groupoid whose vertex group π has a normal abelian subgroup $\mathcal{D}$ (the vertex bundle of a Zappa–Szép factorisation), with quotient $\Gamma = \pi/\mathcal{D}$ the vertex group of the orbit base. Then $H_{\mathrm{cat}}^2(G; A) \cong H^2(\pi; M)$ is the abutment of the Lyndon–Hochschild–Serre spectral sequence $E_2^{p,q} = H^p(\Gamma; H^q(\mathcal{D}; M))$, in which*

- (i) *the base edge $E_2^{2,0} = H^2(\Gamma; M^{\mathcal{D}})$ is the home of the holonomy $[\omega]$;*
- (ii) *the fibre edge $E_2^{0,2} = H^2(\mathcal{D}; M)^\Gamma$ is the additive face — the star-reassembly (Schreier) class, next to which Ferri prunability refines to a degree-one descent datum (§8);*
- (iii) *the transgression $d_2: E_2^{0,1} \rightarrow E_2^{2,0}$ is the holonomy: $d_2(\varphi) = \varphi_*[\omega]$.*

Consequently the direct sum (1) holds if and only if $[\omega] = 0$; whenever $[\omega] \neq 0$ and some φ has $\varphi_[\omega] \neq 0$, the base edge does not survive and no such sum exists. The two obstructions are never simultaneously present as independent summands.*

The one-line reading of Theorem 1: *a nonzero holonomy is not a summand next to prunability — it is the differential that transgresses itself away.* This is the group-theoretic recast Ferri asked

for: prunability is the fibre edge, the holonomy is the transgression, and both are edges of one extension’s spectral sequence.

The theorem describes what happens on each edge in the abstract. To see that the fibre edge is *genuinely occupied* — that prunability is a real, nonzero, computable class and not a formal placeholder — we compute it on Ferri’s own example.

Theorem 2 (Witness, computed, §7). *On Ferri’s $K_{2,3}$ [7, Ex. 4.29] — a connected degree-two groupoid on $\mathbb{Z}/4$ that is non-prunable (Ferri’s Counterexample 5.2) — the two edges decouple: the base edge vanishes, $[\omega] = 0$ (G is connected, so the orbit base is contractible), while the fibre edge carries a nonzero class $[\pi] \in H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$. A prunable Klein-four control kills both edges. Thus the fibre edge is genuinely occupied, decoupled from the holonomy, on Ferri’s actual numbers. What that class is — the degree-two star-reassembly (Schreier) datum — and where Ferri prunability sits relative to it — one degree below, as a descent condition — is pinned down in §8, which sharpens (and in one respect corrects) the naive reading “the fibre edge is prunability”.*

Method. Normality turns the whole question into ordinary group cohomology along the extension $1 \rightarrow \mathcal{D} \rightarrow \pi \rightarrow \Gamma \rightarrow 1$ (§3); the two obstruction theories are then read off as the two axes of the LHS E_2 -page (§4); a one-page cocycle computation identifies the transgression with the extension class (§5); and a survival count gives the collapse locus and the no-go (§6). The $K_{2,3}$ computation (§7) is the additive twin of the multiplicative Schreier line for $[\omega]$: the fibre-edge class is the nonsplitting of the additive extension $0 \rightarrow \{0, 2\} \rightarrow \mathbb{Z}/4 \rightarrow \mathbb{Z}/2 \rightarrow 0$ read at each star, and it is machine-checked in Lean.

Consequences. Three previously separate results — the holonomy face (H), the prunability face (P), and their proved separation (S) [3, 4] — become one object. The separation theorem is recovered as the contractible-base slice $\Gamma \simeq *$, where the base edge is forced to zero and only the fibre edge survives; $K_{2,3}$ is exactly this slice, realised on a real example. For the grant Theory strand this is the honest “internal replacement / Zappa–Szép, fully cohomological” statement: compositional correctness of the directed-container composite is $[\omega] = 0$, which is the collapse of one spectral sequence whose two edges are the base handoff geometry and the additive star-reassembly class (prunability itself the degree-one descent datum below the fibre edge, §8).

A monolithic precursor. The whole object we filter — the categorical abutment $H_{\text{cat}}^2(G)$ — has an independently published skew-brace analogue: Wazzan and Oluyori’s H_{ZS}^2 [8] classifies extensions of the fused Zappa–Szép object as a single group. Our contribution is to *filter* that abutment — to separate its two edge obstructions and identify their coupling as the holonomy (§10).

Outline. §2 fixes the setup and defines $[\omega]$ and prunability. §3 reduces to a group extension. §4 identifies the two edges. §5 proves the transgression is the holonomy. §6 gives the collapse locus, the no-go, and the finite witnesses. §7 computes the fibre edge on $K_{2,3}$ and answers Remark 3.20. §8 sharpens that reading via the descent theorem (prunability is degree-one, the fibre class degree-two, independent). §9 assembles the unification. §10 positions the result against Wazzan–Oluyori’s monolithic H_{ZS}^2 . §11 states the honest scope, including the open $|L_\lambda| \geq 3$ linear reduction.

2 Preliminaries

We keep only what the proofs use. Throughout, G is a connected groupoid with a chosen base object x_0 and vertex group $\pi = G(x_0, x_0)$; A is a G -module and $M := A(x_0)$ its value at x_0 . All cohomology is group cohomology unless decorated. The Lyndon–Hochschild–Serre (LHS) spectral sequence of a group extension and its five-term exact sequence are used as in Weibel [10, §6.8]; the transgression formula in the general twisted case is Neukirch–Schmidt–Wingberg [11].

Zappa–Szép factorisation and the holonomy $[\omega]$. A Zappa–Szép (matched pair) factorisation presents the composite groupoid K as an interaction $\mathcal{C} \bowtie \mathcal{D}$ of a base groupoid $\mathcal{C}$ and a vertex bundle $\mathcal{D}$ that act on each other. The obstruction to the global factorisation existing is a degree-two class, the *composition holonomy* $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ over the orbit base $\mathrm{Sk}_{\mathcal{C}}$ (the skeleton of $\mathcal{C}$), and $K = \mathcal{C} \bowtie \mathcal{D}$ exists iff $[\omega] = 0$ [1, 2]. When $\mathcal{D}$ is normal in π this is the Schreier class of the extension $1 \rightarrow \mathcal{D} \rightarrow \pi \rightarrow \Gamma \rightarrow 1$ with $\Gamma = \mathrm{Sk}_{\mathcal{C}}$, and it is genuinely nonzero — e.g. $[\omega] \neq 0$ for $\mathbb{Z}/4$ over $\mathbb{Z}/2$ [2]. When the loop bundle is *full*, the orbit base is connected, hence contractible, and $[\omega] = 0$ identically [3].

Prunability. For a skew-brace groupoid, the loops over an object x form a set $\mathrm{loops}(x)$ inside the additive group at x . Ferri calls G *prunable* when the loop subgroupoid is a $\rightarrow$ -ideal — concretely, when $\mathrm{loops}(x)$ is an additive subgroup for every x — so that G may be pruned to its loop bundle [7]. The obstruction is additive and lives on the vertex group: the direction-functor cohomology of Ambra–Duvieusart–Montoli [9] reduces, for a connected groupoid, to $H_{\mathrm{Cat}_B}^2(G; A) \cong H^2(\pi; M)$ [4, T1], the additive face that carries prunability [3]. Ferri’s Remark 3.20 asks for the cohomological picture behind this condition; §7 supplies it on his own example.

The separation, and the synthesis it invites. The two faces were proved orthogonal: $[\omega] \perp$ prunability, from a computed witness [3] and structurally from the decomposition $G \cong \mathbf{I}(\mathrm{ob} G) \times B\pi$ in the full-bundle case [4, T2]. Orthogonality is what makes the direct sum (1) tempting. But (1) silently uses one symbol $\mathrm{Sk}_{\mathcal{C}}$ for two different bases:

- In the *connected/full-loop-bundle* regime, $\mathrm{Sk}_{\mathcal{C}} = \mathbf{I}(\mathrm{ob} G)$ is the coarse groupoid on the objects, a disjoint union of contractible components, so $H^{\geq 1}(\mathrm{Sk}_{\mathcal{C}}; -) = 0$ always and the base summand of (1) is identically zero. Here $G \cong \mathbf{I}(\mathrm{ob} G) \times B\pi$ is a genuine product and (1) says nothing but $H_{\mathrm{cat}}^2 = H^2(\pi)$.
- In the *proper-normal-bundle* regime, $\mathrm{Sk}_{\mathcal{C}} = B\Gamma$ is the classifying groupoid of a quotient group and is generally not contractible; this is where $[\omega] \neq 0$. But now G is *not* a product: $(\mathcal{D}, \Gamma, \pi)$ form a non-split *extension*.

So the two regimes never coexist inside one product. The honest object that sees a non-contractible base is a group extension, and its cohomology is governed by LHS, not by Künneth. We now show $[\omega]$ appears there as the transgression.

3 Reduction to a group extension

We work in the normal Zappa–Szép regime of [2], the only regime in which the base carries a nonzero class.

Proposition 3 (Reduction, [2, Prop. 2.1–2.2], [4, T1]). *Let G be a connected groupoid whose vertex group π admits a normal abelian subgroup $\mathcal{D} \trianglelefteq \pi$ (the vertex bundle of a Zappa–Szép factorisation), with quotient $\Gamma = \pi/\mathcal{D}$ the vertex group of the orbit base $\text{Sk}_C = B\Gamma$. Then:*

- (a) $\mathcal{D}$ is a Γ -module via conjugation, and the Zappa–Szép holonomy $[\omega] \in H^2(\text{Sk}_C; \mathcal{D}) = H^2(\Gamma; \mathcal{D})$ is the Schreier class of the extension

$$1 \longrightarrow \mathcal{D} \longrightarrow \pi \xrightarrow{q} \Gamma \longrightarrow 1; \quad (2)$$

- (b) for a G -module A with $M := A(x_0)$, the categorical (direction-functor) cohomology reduces to the vertex group, $H_{\text{cat}}^2(G; A) = H_{\text{Cat}_B}^2(G; A) \cong H^2(\pi; M)$.

Proof. (a) is [2, Prop. 2.1–2.2]: normality gives $\text{Sk}_C = \pi/\mathcal{D} = \Gamma$ and the defect 2-cochain is the Schreier factor set of (2). (b) is [4, T1], $H_{\text{Cat}_B}^n(G, A) \cong H^n(\pi, M)$, whose $n = 2$ instance rests on the reconstructed Schreier correspondence there and elementary homological algebra. $\square$

Proposition 3 moves the whole question into ordinary group cohomology: how does $H^2(\pi; M)$ decompose along (2), and where do the two faces sit? The answer is the LHS spectral sequence.

4 The spectral sequence and its two edges

Theorem 4 (LHS decomposition of the categorical cohomology). *With the data of Proposition 3, $H_{\text{cat}}^\bullet(G; A) \cong H^\bullet(\pi; M)$ is the abutment of the first-quadrant cohomological spectral sequence*

$$E_2^{p,q} = H^p(\Gamma; H^q(\mathcal{D}; M)) \implies H^{p+q}(\pi; M), \quad (3)$$

of the extension (2) [10, §6.8]. Its two degree-two edge terms are exactly the two obstruction faces:

$$\begin{aligned} E_2^{2,0} &= H^2(\Gamma; M^\mathcal{D}) \quad (\text{base edge: the home of } [\omega]), \\ E_2^{0,2} &= H^2(\mathcal{D}; M)^\Gamma \quad (\text{fibre edge: the additive face}). \end{aligned}$$

The abutment is filtered with associated graded $\text{gr } H^2(\pi; M) = E_\infty^{2,0} \oplus E_\infty^{1,1} \oplus E_\infty^{0,2}$, where $E_\infty^{p,q}$ is the subquotient of $E_2^{p,q}$ surviving the differentials.

Proof. The LHS spectral sequence of a group extension with abelian coefficient module M is standard [10, Thm. 6.8.2]; (3) is its E_2 -page. The edge identifications $E_2^{p,0} = H^p(\Gamma; M^\mathcal{D})$ and $E_2^{0,q} = H^q(\mathcal{D}; M)^\Gamma$ are the definition of the two axes. That the base edge is where $[\omega]$ lives is Proposition 3(a) ($[\omega] \in H^2(\Gamma; \mathcal{D})$, mapped to $H^2(\Gamma; M^\mathcal{D})$ by any Γ -map $\mathcal{D} \rightarrow M^\mathcal{D}$); that the fibre edge is the additive/vertex face is [3, 4]. The three-step filtration in total degree 2 is the standard convergence of a first-quadrant sequence. $\square$

The two faces are not *a priori* summands: they are two edges of one page, and whether they survive to the abutment is decided by the differentials. The relevant one is the transgression, and it is the holonomy itself.

5 The transgression is the holonomy

Lemma 5 (The transgression is cup with the holonomy). *In the five-term exact sequence of (3),*

$$0 \rightarrow H^1(\Gamma; M^{\mathcal{D}}) \xrightarrow{\inf} H^1(\pi; M) \xrightarrow{\text{res}} H^1(\mathcal{D}; M)^{\Gamma} \xrightarrow{\text{tg}} H^2(\Gamma; M^{\mathcal{D}}) \xrightarrow{\inf} H^2(\pi; M),$$

the transgression $\text{tg} = d_2^{0,1}$ is given by pushforward of the extension class: for a Γ -equivariant homomorphism $\varphi \in \text{Hom}_{\Gamma}(\mathcal{D}, M) = H^1(\mathcal{D}; M)^{\Gamma}$,

$$\text{tg}(\varphi) = \varphi_*[\omega] \in H^2(\Gamma; M^{\mathcal{D}}),$$

where $\varphi_: H^2(\Gamma; \mathcal{D}) \rightarrow H^2(\Gamma; M^{\mathcal{D}})$ is induced by φ and $[\omega]$ is the Schreier class of (2).*

Proof. We give the cocycle computation for the case M a trivial π -module (which covers all witnesses below and the $\mathbb{Z}/4$ -Schreier line); the general Γ -module case is [11, transgression formula], identical after keeping the covariant end. Choose a normalised set section $s: \Gamma \rightarrow \pi$ of q , $s(1) = 1$, with Schreier factor set

$$s(a)s(b) = \omega(a, b)s(ab), \quad \omega(a, b) \in \mathcal{D}, \quad [\omega] \in H^2(\Gamma; \mathcal{D}).$$

An element of $E_2^{0,1} = H^1(\mathcal{D}; M)^{\Gamma}$ is a Γ -invariant homomorphism $\varphi: \mathcal{D} \rightarrow M$. Running the construction of the transgression in the double complex of the extension [10, §6.8]: φ lifts to the 1-cochain $\tilde{\varphi} \in C^1(\pi; M)$, $\tilde{\varphi}(g) = \varphi(\mathcal{D}\text{-part of } g)$ along the section, and its image under the vertical differential, pushed to the base, is the 2-cochain

$$(a, b) \mapsto \varphi(\omega(a, b)) = (\varphi_*\omega)(a, b).$$

Thus $\text{tg}(\varphi) = [\varphi_*\omega] = \varphi_*[\omega]$. (Concretely: $\text{tg}(\varphi)$ is the obstruction to extending φ from $\mathcal{D}$ to a homomorphism $\pi \rightarrow M$; it is 0 iff the pushed-out extension $1 \rightarrow M \rightarrow \varphi_*\pi \rightarrow \Gamma \rightarrow 1$ splits, i.e. iff $\varphi_*[\omega] = 0$.) $\square$

6 Collapse and the no-go

Corollary 6 (When the clean sum holds). *The naive direct sum (1) of the two edges equals $H^2(\pi; M)$ whenever (3) collapses at E_2 in total degree 2 with vanishing cross term $E_2^{1,1}$. Each of the following is sufficient:*

- (i) **Coarse/connected base** $\Gamma = 1$: *then $E_2^{p,q} = 0$ for $p > 0$, the base edge and cross term vanish, and $H^2(\pi; M) = E_2^{0,2} = H^2(\mathcal{D}; M)$. Since here $\mathcal{D} = \pi$ this is the identity $H_{\text{cat}}^2(G) = H^2(\pi)$ — the clean sum degenerates to its fibre summand, and $[\omega] = 0$.*
- (ii) **Direct product** $\pi = \mathcal{D} \times \Gamma$: *then $[\omega] = 0$ and the sequence collapses (Künneth), giving the full associated graded $H^2(\pi; M) \cong E_2^{2,0} \oplus E_2^{1,1} \oplus E_2^{0,2}$ — the honest Künneth form, including the cross term $H^1(\Gamma) \otimes H^1(\mathcal{D})$ that (1) omits.*
- (iii) **Coprime orders** $\gcd(|\mathcal{D}|, |\Gamma|) = 1$ (Schur-Zassenhaus): *$E_2^{p,q} = 0$ for $p, q > 0$, so $H^2(\pi; M) = E_2^{2,0} \oplus E_2^{0,2}$ and $[\omega] = 0$.*

In every collapsing case the base class is forced to zero: $[\omega] = 0$.

Proof. (i) A coarse groupoid is equivalent to the terminal category, so $H^{\geq 1}(\Gamma; -) = 0$; the surviving column is $p = 0$. (ii) A direct product splits (2), so $[\omega] = 0$; collapse is the Eilenberg–Zilber/Künneth theorem for $B\mathcal{D} \times B\Gamma$. (iii) For coprime orders the higher cohomology of each factor is annihilated by both $|\mathcal{D}|$ and $|\Gamma|$, hence 0 off the edges; a split extension (Schur–Zassenhaus) has $[\omega] = 0$. $\square$

Corollary 6 is the precise content of (1): it is *true exactly on the locus* $[\omega] = 0$, and there it is either the trivial statement $H_{\text{cat}}^2 = H^2(\pi)$ (connected case, the prunability-only corner realised by $K_{2,3}$ in §7) or the genuine Künneth sum (direct product) — but in neither case is the base summand nonzero. The interesting “both nonzero” regime is empty.

Theorem 7 (No clean sum when $[\omega] \neq 0$). *Suppose $[\omega] \neq 0$ and there is a Γ -equivariant $\varphi: \mathcal{D} \rightarrow M$ with $\varphi_*[\omega] \neq 0$ in $H^2(\Gamma; M^{\mathcal{D}})$ (e.g. $M = \mathcal{D}$, $\varphi = \text{id}$, whenever $[\omega] \neq 0$). Then:*

- (a) *the transgression $d_2^{0,1} = \text{tg}$ is nonzero (Lemma 5), so the base edge does not survive: $E_{\infty}^{2,0} \subsetneq E_2^{2,0}$, and the image $\varphi_*[\omega]$ of the holonomy is killed in the abutment;*
- (b) *consequently $H^2(\pi; M)$ is not the direct sum (1) of the two edges: $\dim H^2(\pi; M) < \dim E_2^{2,0} + \dim E_2^{1,1} + \dim E_2^{0,2}$ whenever $\text{tg} \neq 0$.*

In particular the two faces are never simultaneously present as independent summands of $H_{\text{cat}}^2(G)$: a nonzero base class is not a summand but a differential.

Proof. (a) By Lemma 5, $\text{tg}(\varphi) = \varphi_*[\omega] \neq 0$, so $d_2^{0,1} \neq 0$. The differential $d_2^{0,1}: E_2^{0,1} \rightarrow E_2^{2,0}$ has nonzero image, hence $E_3^{2,0} = E_2^{2,0} / \text{im } d_2^{0,1}$ is a proper quotient; as the base edge receives no further differentials ($d_r^{p,0}$ into $(2,0)$ come from $(2-r, r-1)$, a negative column for $r \geq 3$), $E_{\infty}^{2,0} = E_3^{2,0} \subsetneq E_2^{2,0}$, and the class $\varphi_*[\omega]$ (a generator of $\text{im } d_2$) is annihilated. (b) The abutment has $\dim H^2 = \sum_{p+q=2} \dim E_{\infty}^{p,q} \leq \sum_{p+q=2} \dim E_2^{p,q}$ with strict inequality once any $E_{\infty}^{p,q} \subsetneq E_2^{p,q}$; part (a) gives this at $(2,0)$. $\square$

Corollary 8 (Finite witnesses, machine-checked). *For trivial $\mathbb{Z}/2$ -coefficients $M = \mathbb{F}_2$ (2026-09-17-`lhs-decomposition-check.py`):*

$\pi = \mathcal{D}.\Gamma$	<i>split?</i>	$[\omega]$	$\dim H^2(\pi)$	$\dim(E_2^{2,0} \oplus E_2^{1,1} \oplus E_2^{0,2})$	rk tg	<i>verdict</i>
$\mathbb{Z}/4 = \mathbb{Z}/2.\mathbb{Z}/2$	<i>no</i>	$\neq 0$	1	$1+1+1 = 3$	1	no sum (base edge dies)
$Q_8 = (\mathbb{Z}/2)^2.\mathbb{Z}/2$	<i>no</i>	$\neq 0$	2	$3+2+1 = 6$	1	no sum
$\mathbb{Z}/2 \times \mathbb{Z}/2 = \mathbb{Z}/2.\mathbb{Z}/2$	<i>yes</i>	0	3	$1+1+1 = 3$	0	<i>clean</i> (full Künneth)
$\mathbb{Z}/6 = \mathbb{Z}/2.\mathbb{Z}/3$	<i>yes</i>	0	1	$0+0+1 = 1$	0	<i>clean</i> (fibre only)

For $\mathbb{Z}/4$ the transgression is an isomorphism $E_2^{0,1} \xrightarrow{\sim} E_2^{2,0}$, so the entire base edge $H^2(\mathbb{Z}/2; \mathbb{F}_2)$ — the home of the nonzero Schreier $[\omega]$ of $\mathbb{Z}/4$ over $\mathbb{Z}/2$ [2] — vanishes from the abutment. The naive sum overcounts by 2; the true $H^2(\mathbb{Z}/4; \mathbb{F}_2)$ is 1-dimensional.

7 The fibre edge is occupied: Ferri’s $K_{2,3}$

Theorem 7 shows the base edge cannot be an independent summand. It says nothing about whether the fibre edge is ever nonzero on a real skew brace — whether prunability is an actual class or a formal slot. We settle this on Ferri’s own example, which is *connected* (so we are in the collapse corner 6(i), $[\omega] = 0$) yet *non-prunable*. This is the honest place to test prunability: it is the corner where the holonomy is forced to zero, so any obstruction that remains lives on the fibre edge alone.

The example. Ferri’s $K_{2,3}$ [7, Ex. 4.29, Table 3] has underlying set $A = \mathbb{Z}/4$ and two objects S_2, S_3 ; the direction data give a connected degree-two groupoid. Its loop bundle is

$$\text{loops}(S_2) = \{0, 3\}, \quad \text{loops}(S_3) = \{0, 1\}.$$

Neither is an additive subgroup of $(\mathbb{Z}/4, +)$ ($3 + 3 = 2 \notin \{0, 3\}$, $1 + 1 = 2 \notin \{0, 1\}$), so G is *not* prunable — this is Ferri’s Counterexample 5.2.

Remark 9 (The quasigroup subtlety, stated plainly). The star operations $\bullet_{S_2}, \bullet_{S_3}$ of Table 3 are left *quasigroups*, not groups: their rows are permutations but their columns are not, and both are non-commutative and non-associative (machine-checked: e.g. $\bullet_{S_2}$ has $(1 \bullet 1) \bullet 1 = 1 \neq 3 = 1 \bullet (1 \bullet 1)$). Since every group of order 4 is abelian, $\bullet_\lambda$ cannot itself be the additive group. The skew brace’s genuine additive group on A is the ordinary $(\mathbb{Z}/4, +)$ (Ferri’s convention $a \cdot b = a + b$), and it is with respect to $(\mathbb{Z}/4, +)$ that “additive subgroup”, hence prunability, is meaningful. The loop sets *are* closed under the quasigroup $\bullet_\lambda$ (they are sub-left-quasigroups), but that is not the condition Ferri’s prunability tests; the defect is precisely that they are not subgroups of $(\mathbb{Z}/4, +)$. We flag this because the identification of the fibre-edge class below with prunability rests on this $(\mathbb{Z}/4, +)$ reading (see §11).

Base edge $E_2^{2,0} = H^2(\Gamma; M^{\mathcal{D}})$. $K_{2,3}$ is connected, so the orbit base is the coarse groupoid on $\{S_2, S_3\}$, which is contractible: $H^{\geq 1}(\Gamma; -) = 0$ for any coefficients. Hence $[\omega] = 0$. This is rigorous and coefficient-independent (Corollary 6(i) on this example).

Fibre edge $E_2^{0,2} = H^2(\mathcal{D}; M)^\Gamma$. With $\Gamma \simeq *$ the Γ -invariants are everything, and the fibre edge is $H^2(\mathcal{D}; M)$ computed at each star. Read as the additive Schreier class of the extension

$$0 \longrightarrow A_2 \longrightarrow (\mathbb{Z}/4, +) \longrightarrow \mathbb{Z}/2 \longrightarrow 0, \quad A_2 = \{0, 2\} \cong \mathbb{Z}/2,$$

we have $H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$, and the extension is *nonsplit* at both S_2 and S_3 : its Schreier cocycle is the carry $f(1, 1) = 2 \in A_2$, which is not a coboundary. Hence the fibre-edge class

$$[\pi]_{S_2} = [\pi]_{S_3} = \text{the nonzero generator of } H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2.$$

This computation is machine-checked in Lean, sorry-free (2026-09-17-fibre-h2-z2.lean): $H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$ as the cokernel of the norm; the $\mathbb{Z}/4$ carry cocycle $f(a, b) = a \wedge b$ is a genuine cocycle and *not* a coboundary (matching at $(0, 0)$ forces $g(0) = 0$, whence $\delta^1 g(1, 1) = 0 \neq 1$); and the Klein control cocycle is 0.

The control. Replacing the $\mathbb{Z}/4$ star by the Klein four-group V_4 keeps the groupoid connected but makes the loops an additive subgroup, so G becomes prunable. Now the additive extension $0 \rightarrow \mathbb{Z}/2 \rightarrow V_4 \rightarrow \mathbb{Z}/2 \rightarrow 0$ *splits*, so the fibre-edge class is zero. Both edges are dead. The control isolates the fibre edge as the genuine detector of prunability: it is nonzero exactly when G fails to be prunable.

case	prunable?	base H^2	$[\omega]$	star extension	$[\pi]$ (fibre edge)
Ferri $K_{2,3}$ ($\mathbb{Z}/4$ stars)	no	0	0	nonsplit (both S_2, S_3)	$\neq 0$ (both)
Klein control (V_4 star)	yes	0	0	split	0

Answer to Remark 3.20. On Ferri’s own $K_{2,3}$, the prunability obstruction *is* the fibre-edge class $[\pi] \in H^2(\mathcal{D}; M)^\Gamma$ of the LHS spectral sequence (3), equivalently the nonsplitting of the additive star extension $0 \rightarrow A_2 \rightarrow (\mathbb{Z}/4, +) \rightarrow \mathbb{Z}/2 \rightarrow 0$. This is the additive twin of the multiplicative Schreier line that carries $[\omega]$: $[\omega]$ is the nonsplitting of the multiplicative extension $1 \rightarrow \mathcal{D} \rightarrow \pi \rightarrow \Gamma \rightarrow 1$ on the base edge, and prunability is the nonsplitting of the additive extension on the fibre edge. The dividing line, $\mathbb{Z}/4$ (nonsplit) versus Klein (split), is exactly the additive twin of the $\mathbb{Z}/4$ -versus- $\mathbb{Z}/2 \times \mathbb{Z}/2$ line for $[\omega]$ in Corollary 8. This is the additive twin the fibre edge carries; §8 says precisely what that twin *is*, and where Ferri prunability sits relative to it.

8 What the fibre edge really is: prunability is one degree lower

The witness of §7 shows the fibre edge is occupied by a nonzero H^2 class on $K_{2,3}$, and it is tempting to read that class *as* prunability. In the untwisted core — the regime where the loop bundle satisfies Ferri’s subgroup condition **SUB**, so that “prune to the loops” is even well posed — the identification must be refined: prunability there is a *degree-one* descent datum, and the degree-two fibre-edge class is a logically independent *star-reassembly* datum. We record the three facts that pin this down, proved in [6]; they sharpen, and in one respect correct, the reading of Theorem 2.

Setup (untwisted core). Take abelian stars $(\text{St } \lambda, +_\lambda)$ with left action $a \mapsto (-)$, loop bundle $G^{\text{loops}} = \bigsqcup_\lambda L_\lambda$, and Ferri’s conditions **SUB** (each L_λ an additive subgroup of its star) and

$$(9) \quad a \mapsto L_{\text{tg}(a)} \subseteq L_{\text{s}(a)} \quad \text{for every arrow } a.$$

Complete prunability (the loop bundle a $\mapsto$ -ideal) is **SUB** $\wedge$ (9). Write $\mathcal{A}: \lambda \mapsto (\text{St } \lambda, +_\lambda)$, $\mathcal{A}(a) := a \mapsto (-)$, for the star local system (a functor to abelian groups; each $\mathcal{A}(a)$ an isomorphism, [5]).

Theorem 10 (Loop invariance is automatic [6, Thm. A]). *Under **SUB**, condition (9) restricted to loop arrows holds automatically: for every loop $\ell \in L_\lambda$, $\ell \mapsto L_\lambda = L_\lambda$. Prunability can therefore fail only through cross-object transport.*

Theorem 11 (Prunability is descent, not a Schreier class [6, Thm. B]). *Under **SUB**, the following are equivalent: (i) (9) holds (G^{loops} is prunable); (ii) $\{L_\lambda\}$ is a $\mapsto$ -subfunctor (flat section) of $\mathcal{A}$, i.e. $a \mapsto L_\mu = L_\lambda$ for every $a: \lambda \rightarrow \mu$; (iii) the action descends to the quotients $\mathcal{A}(\lambda)/L_\lambda$, making $G^{\text{loops}} = \ker(\mathcal{A} \twoheadrightarrow \mathcal{A}/\mathcal{L})$ a kernel. For $|L_\lambda| = 2$ this linearises to a degree-one cochain $\delta: \text{ob } G \rightarrow A_0/L_0$ with (9) $\iff \delta \equiv 0$. Thus prunability is a first-order descent (parallel-transport) obstruction, one degree below a Schreier extension class.*

Theorem 12 (The Schreier class is blind to prunability [6, Thm. C]). *Let $e_\lambda = [0 \rightarrow L_\lambda \rightarrow (\text{St } \lambda, +_\lambda) \rightarrow \text{St } \lambda / L_\lambda \rightarrow 0] \in H^2(\text{St } \lambda / L_\lambda; L_\lambda)$ be the Schreier class of the star extension — the degree-two datum the fibre edge carries. Over the star $\mathbb{Z}/4 \times \mathbb{Z}/2$, all four combinations of (e split/nonsplit) $\times$ ((9) holds/fails) occur among **SUB** skew braces:*

	<i>e split</i>	<i>e nonsplit</i>
(9) holds (prunable)	96	48
(9) fails (non-prunable)	192	96

In particular there are 48 prunable skew braces with a nonsplit star and 192 non-prunable ones with a split star. Hence e neither implies nor is implied by prunability: the degree-two fibre class does not detect (9).

Corollary 13 (Two independent data on the loop bundle). *In the untwisted core the loop bundle carries two logically independent obstructions:*

- (1) *the descent/ideal datum (Theorem 11) — a degree-one flat-section obstruction, vanishing iff G^{loops} is a kernel; this is prunability; and*
- (2) *the reassembly datum — the degree-two star Schreier class e on the fibre edge $E_2^{0,2}$, and multiplicatively the holonomy $[\omega]$ on the base edge $E_2^{2,0}$ — governing how a total object is reassembled from kernel and quotient.*

By Theorem 12 these are uncorrelated. The nonzero fibre-edge class of §7 is datum (2), not (1).

Reconciliation with $K_{2,3}$. The $K_{2,3}$ witness is unaffected as a *computation* — the fibre edge does carry the nonzero generator of $H^2(\mathbb{Z}/2; \mathbb{Z}/2)$ — but its *reading* is now precise. On $K_{2,3}$ **SUB** fails (the loops $\{0, 3\} \subset \mathbb{Z}/4$ are not an additive subgroup), so the descent picture of Theorem 11 does not apply and the nonzero fibre class measures the ambient star’s non-splitness — datum (2) — which there co-occurs with non-prunability only because **SUB** fails. In the **SUB** regime, Theorem 12 shows the two part company in every cell. So the honest answer to Ferri’s Remark 3.20 is sharper than “prunability is the fibre-edge H^2 ”: prunability is the *degree-one* descent condition that G^{loops} be a kernel (Theorem 11), and the degree-two fibre-edge class is a distinct, independent reassembly datum.

An ideal is a kernel (degree-one descent); an extension is a reassembly (degree-two H^2). The loop bundle carries both, and they are independent.

Honest scope. Theorem 11’s equivalences are general, but the clean degree-one cochain δ uses $|L_\lambda| = 2$; for $|L_\lambda| \geq 3$ the descent obstruction is a section of a bundle of order- n subgroups (a torsor-valued first cohomology) and its reduction to a single linear class is **open**. The reconciliation above is for the untwisted (abelian-star, **SUB**) core; the fully dynamical (quasigroup-only) case is as in §11.

9 What the cluster is, correctly

The three cluster results unify into one object — a spectral sequence, not a direct sum.

Theorem 14 (Structural unification). *Under Proposition 3, the categorical cohomology $H_{\text{cat}}^2(G; A) \cong H^2(\pi; M)$ is the total degree-two term of the LHS spectral sequence (3), in which:*

1. *the fibre edge $E_2^{0,2} = H^2(\mathcal{D}; M)^\Gamma$ is the additive face (P) — the star-reassembly (Schreier) class, occupied by a nonzero class on Ferri’s $K_{2,3}$ (§7); Ferri prunability proper is the degree-one descent datum below it (§8, independent by Theorem 12);*
2. *the base edge $E_2^{2,0} = H^2(\Gamma; M^\mathcal{D})$ is the holonomy face, receiving the transgression d_2 ;*
3. *the transgression $d_2: E_2^{0,1} \rightarrow E_2^{2,0}$ is the Zappa–Szép holonomy, $d_2(\varphi) = \varphi_*[\omega]$ (H).*

*The proved separation $[\omega] \perp \text{prunability (S)}$ is the special case $\Gamma \simeq *$ (base contractible, only the fibre edge survives) — precisely the corner $K_{2,3}$ inhabits. In general the two faces are not orthogonal summands but the two axes of one page, coupled by the transgression, which is $[\omega]$ itself. The clean direct sum (1) holds iff this coupling vanishes, i.e. iff $[\omega] = 0$.*

Proof. Assemble Theorem 4 (edges), Lemma 5 (transgression = holonomy), Corollary 6 (collapse $\Leftrightarrow$ the true locus of (1)), Theorem 7 (no-go off that locus), and §7 (the fibre edge is occupied). The separation is Corollary 6(i): $\Gamma = 1 \Rightarrow$ base edge and cross term vanish, $H_{\text{cat}}^2 = H^2(\mathcal{D}) = H^2(\pi)$, orthogonality is the triviality of the base axis. $\square$

Remark 15 (Why this is the right correction). The conjecture (1) failed for a structural reason worth stating plainly: a nonzero base class $[\omega]$ is a non-split extension, and a non-split extension is exactly a nonzero transgression. One cannot “add” the holonomy as a summand next to the prunability class, because the holonomy is the mechanism by which the base cohomology is glued into the fibre cohomology. The separation theorem saw the two faces as orthogonal only because it lived on the contractible-base slice where the transgression is forced to zero; off that slice, orthogonality is replaced by a single coupling differential. This is strictly more informative, and it is the honest “internal replacement / Zappa–Szép” deliverable: compositional correctness of the directed-container composite $= [\omega] = 0 =$ the collapse of one spectral sequence, whose two edges are the base handoff geometry and the vertex/additive prunability.

10 Related work: the monolithic skew-brace abutment

The whole object we have filtered — the categorical abutment $H_{\text{cat}}^2(G) \cong H^2(\pi; M)$ — has, on the skew-brace side, an independently published analogue. Wazzan and Oluyori [8] define a second cohomology $H_{ZS}^2(A; M)$ for a Zappa–Szép skew brace $A = Y \bowtie K$ (a “digroup” base carrying two compatible group operations) and prove it classifies extensions $1 \rightarrow M \rightarrow E \rightarrow A \rightarrow 1$ of the *whole* fused object A ; their lifting corollary places the obstruction to promoting a semidirect skew brace to a fully mutual (Zappa–Szép) one inside H_{ZS}^2 . This is the same narrative move as ours — a degree-two class detecting when two interacting factors assemble into one — carried out for skew braces rather than for the groupoid G .

H_{ZS}^2 is the abutment, not an edge (computed placement). H_{ZS}^2 is the skew-brace analogue of our *whole* abutment $H_{\text{cat}}^2(G)$, presented monolithically; it is neither the base edge $E_2^{2,0}$ that carries $[\omega]$ nor the fibre edge $E_2^{0,2}$ that carries the additive class. Two features fix this reading. First, H_{ZS}^2 classifies extensions of the *total* fused object $A = Y \bowtie K$, not of a single factor — exactly the abutment’s job, whereas each of our two edges sees only one factor of the extension (2). Second, as reported (see the provenance note below), its cochains split into a *pair*, one component valued in each of the two factors Y, K , so that the additive (fibre / prunability) and multiplicative (base / $[\omega]$) directions are fused at once rather than separated by degree. Simultaneous fusion of both directions is the signature of a total-degree abutment, not of either axis of a spectral sequence.

What the present filtration adds. None of the spectral-sequence structure of this paper appears in [8]: there is no filtration of H_{ZS}^2 , no base/fibre edge, no transgression, and no direct-sum question. Our results are precisely the refinement that a monolithic abutment invites. The Lyndon–Hochschild–Serre spectral sequence (3) filters the whole cohomology; Theorem 4 identifies its two degree-two edges as the two obstruction faces; Lemma 5 identifies the transgression with the holonomy, $d_2(\varphi) = \varphi_*[\omega]$; and Theorem 7 proves the two are never independent summands. Read against [8], the contribution is exactly: *filter the monolithic abutment into its two obstruction edges, and identify the coupling between them as the holonomy*. We work over $\mathbf{DCont} \simeq \mathbf{Cat}$ rather than over skew braces, so the two theories share a narrative and a degree, not a category; whether the LHS

filtration transports verbatim to H_{ZS}^2 — constructing a spectral sequence of the digroup extension whose two edges are literally $[\omega]$ and prunability — is a natural question we do not settle here.

Provenance and scope of this comparison. Our reading of [8] is from a reader-proxy of the journal page (a direct fetch is refused), so the placement above is *computed from that reading* rather than proved, and we deliberately do not reproduce the cochain differentials verbatim; the citation, the author list, and the finer definitional details should be verified against the published version. The claim we actually rely on — that H_{ZS}^2 classifies extensions of the whole A and carries the semidirect-to-Zappa–Szép lifting obstruction — is the paper’s stated main cohomological result and is robust to transcription. At the coarse narrative level “a Zappa–Szép composite carries a degree-two lifting obstruction” the two works overlap; the spectral sequence, the edge placement, the transgression identification, and the $[\omega] \perp$ prunability separation are new here.

11 Status and honest scope

Proved. Theorem 4 (LHS decomposition; standard SS + proved edge identifications), Lemma 5 (transgression = holonomy; self-contained cocycle proof for trivial coefficients, cited [11] in general), Corollary 6 (collapse locus = $\{[\omega] = 0\}$), Theorem 7 (no-go), Theorem 14 (unification). These rest on the proved Proposition 3(a)(b) = [2, 4], the classical LHS machinery [10, 11], and Schur–Zassenhaus.

Computed. The four finite witnesses of Corollary 8 (machine-checked bar complex over $\mathbb{F}_2$ and five-term rank count) and the $K_{2,3}$ decoupling table of §7. The fibre-edge kernel ($H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$, $\mathbb{Z}/4$ carry cocycle $\neq 0$ = Klein) is machine-checked in Lean, sorry-free (2026-09-17-fibre-h2-z2.lean).

Not claimed.

- The clean direct sum (1) with both summands nonzero — **refuted** (Theorem 7).
- **The fibre-edge identification, resolved and corrected in the SUB core** (§8). The naive reading “prunability *is* the degree-two fibre-edge H^2 ” is *false* in the untwisted (abelian-star, **SUB**) core: there prunability is a *degree-one* descent datum (Theorem 11), logically independent of the fibre edge’s degree-two Schreier class (Theorem 12, all four $\mathbb{Z}/4 \times \mathbb{Z}/2$ cells). The $K_{2,3}$ witness’s nonzero fibre class measures star-nonsplitness (datum (2)), co-occurring with non-prunability only because **SUB** fails there. What remains **open** is the *degree-one* reduction: for loop groups $|L_\lambda| \geq 3$ the descent obstruction is a torsor-valued first cohomology (Theorem 11’s clean cochain δ uses $|L_\lambda| = 2$), and a single linear class realising it is not established. The twisted (quasigroup-only, non-trivial Γ -action) case also remains open; on $K_{2,3}$ the reading rests on $+\lambda = (\mathbb{Z}/4, +)$ (Ferri’s Counterexample 5.2 convention; Remark 9). This supersedes the earlier tracked node **prunability-is-fibre-edge-class**, now resolved as “descent, not H^2 ”.
- The nonabelian / non-normal Zappa–Szép matched-pair regime (Kac/Pirashvili H^2) is out of scope, as in [2].

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